10 Fig. 10 shows the graph of $y = (k-x)\ln x$ where k is a constant ($k > 1$).

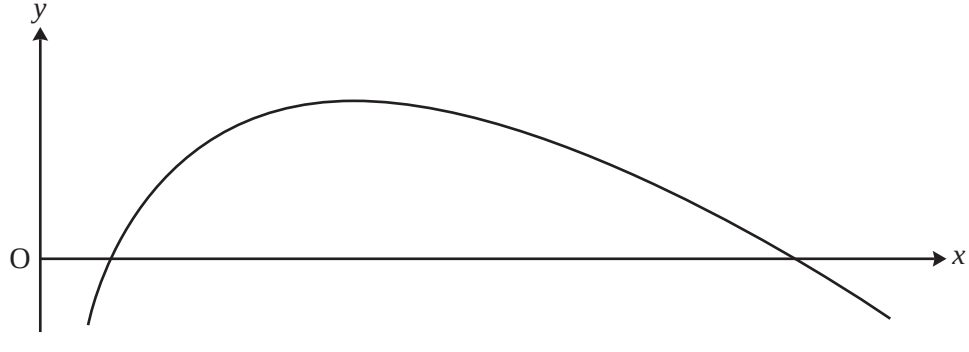

Fig. 10

Find, in terms of k, the area of the finite region between the curve and the x-axis. **[8]**